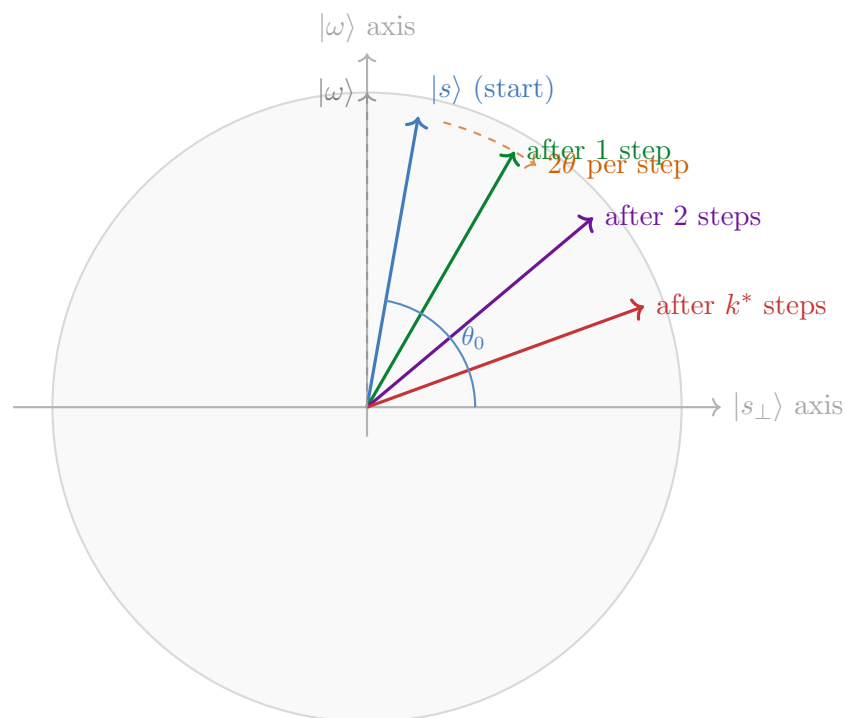

Grover's Quantum Search Algorithm

Lecture Notes — Quantum Computing
From Oracle to Quadratic Speedup: Geometry and Algebra



Prerequisites: Quantum states, Dirac notation, single- and multi-qubit gates, tensor products, measurement postulate.

Goals: Understand the oracle model; derive the Grover iterate geometrically and algebraically; prove the $\mathcal{O}(\sqrt{N})$ speedup; work through complete examples.

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1 Motivation: Why Quantum Search?

1.1 The Classical Unordered Search Problem

Suppose you have a list of N items in *no particular order*. You are given a black-box function that can check whether a given item is the one you seek. How many checks do you need?

- **Best case:** 1 (you get lucky on the first try).
- **Worst case:** N (the target is the last item checked).
- **Average case:** $N/2$.

Without any structure in the data (no sorting, no hash table), there is *no classical shortcut*. The lower bound is provably $\Omega(N)$.

1.2 The Quantum Advantage

Quantum mechanics allows us to query all N items *simultaneously* via superposition, and then use **amplitude amplification** to make the target item stand out. The result, due to Lov Grover (1996), is:

Theorem: Grover's Main Result

An unordered database of N items can be searched using $\mathcal{O}(\sqrt{N})$ quantum queries. This is a **quadratic speedup** over the classical $\mathcal{O}(N)$, and it is **provably optimal** — no quantum algorithm can do better.

Note

A quadratic speedup may sound modest, but it is enormous in practice. For $N = 10^{12}$ (one trillion items), classical search needs $\sim 10^{12}$ steps, while Grover needs only $\sim 10^6$ steps — a *million-fold* reduction.

2 Problem Setup and Oracle Model

2.1 Formal Problem Statement

Definition: Unstructured Search

Let $n \in \mathbb{N}$ and $N = 2^n$. We are given a function

$$f : \{0, 1\}^n \longrightarrow \{0, 1\}$$

implemented as a quantum oracle (black box). There exists a unique **target** $\omega \in \{0, 1\}^n$ such that

$$f(x) = \begin{cases} 1 & \text{if } x = \omega \\ 0 & \text{if } x \neq \omega \end{cases}$$

Goal: find ω by querying f as few times as possible.

We work in the n -qubit Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ with the computational basis $\{|x\rangle : x \in \{0, 1\}^n\}$, so $\dim \mathcal{H} = N$.

2.2 The Standard Oracle

The standard reversible oracle is a unitary on $n + 1$ qubits:

$$O_f : |x\rangle |b\rangle \mapsto |x\rangle |b \oplus f(x)\rangle, \quad x \in \{0, 1\}^n, b \in \{0, 1\}$$

The top n qubits are the **query register**; the bottom qubit is the **target/ancilla register**.

2.3 The Phase Oracle via Kick-Back

We derive the **phase oracle** from O_f by the *phase kick-back trick*. Initialise the ancilla in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$\begin{aligned} O_f |x\rangle |-\rangle &= O_f |x\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}(|x\rangle |f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle) \\ &= \frac{1}{\sqrt{2}}|x\rangle (|f(x)\rangle - |1 \oplus f(x)\rangle) \\ &= (-1)^{f(x)} |x\rangle |-\rangle \end{aligned} \tag{1}$$

Since the ancilla is unchanged, we can ignore it and define the **phase oracle** acting on n qubits:

Definition: Phase Oracle U_f

$$U_f |x\rangle = (-1)^{f(x)} |x\rangle = \begin{cases} -|x\rangle & \text{if } x = \omega \\ |x\rangle & \text{if } x \neq \omega \end{cases}$$

Equivalently, as an outer-product formula:

$$U_f = I - 2|\omega\rangle\langle\omega|$$

Verification:

$$(I - 2|\omega\rangle\langle\omega|) |x\rangle = |x\rangle - 2 \underbrace{\langle\omega|x\rangle}_{=\delta_{x\omega}} |\omega\rangle = \begin{cases} -|\omega\rangle & x = \omega \\ |x\rangle & x \neq \omega \end{cases} \checkmark$$

2.4 Oracle Circuit Diagram

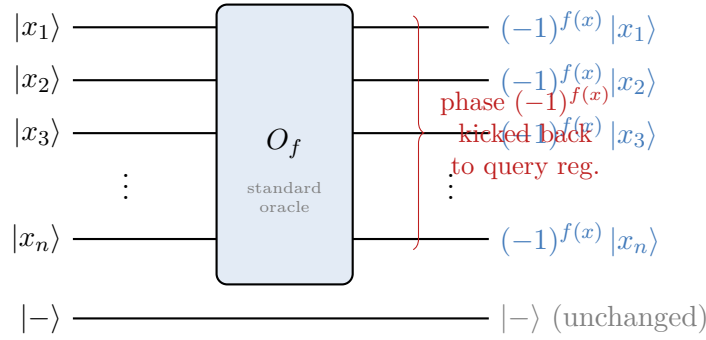


Figure 1: Standard oracle O_f with ancilla $|-\rangle$. The phase $(-1)^{f(x)}$ is “kicked back” onto the query register, leaving the ancilla unchanged. This implements the phase oracle U_f .

3 Initial State and Key Operators

3.1 Uniform Superposition

Apply the n -qubit Hadamard transform $H^{\otimes n}$ to $|0\rangle^{\otimes n}$:

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \underbrace{\sum_{x=0}^{N-1} |x\rangle}_{|s\rangle}$$

Every basis state has equal amplitude $1/\sqrt{N}$. Key inner products:

$$\langle s|\omega\rangle = \frac{1}{\sqrt{N}}, \quad \langle s|s\rangle = 1, \quad \langle \omega|\omega\rangle = 1.$$

3.2 The Two Grover Operations

3.2.1 Operation 1 — Oracle: Reflect about $|\omega\rangle^\perp$

$$U_f = I - 2|\omega\rangle\langle\omega|$$

Geometrically: U_f is a **reflection of the state vector about the hyperplane perpendicular to $|\omega\rangle$** . It flips the sign of the $|\omega\rangle$ component and leaves everything orthogonal to $|\omega\rangle$ unchanged.

3.2.2 Operation 2 — Diffusion: Reflect about $|s\rangle$

$$D = 2|s\rangle\langle s| - I$$

In the computational basis, D can be implemented as $D = H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n}$, which gives:

$$D|x\rangle = (2|s\rangle\langle s| - I)|x\rangle = 2\langle s|x\rangle|s\rangle - |x\rangle = \frac{2}{N} \sum_y |y\rangle - |x\rangle \quad (2)$$

If the current state is $|\psi\rangle = \sum_x \alpha_x |x\rangle$, then after D :

$$\alpha_x \mapsto 2\bar{\alpha} - \alpha_x, \quad \bar{\alpha} = \frac{1}{N} \sum_{x=0}^{N-1} \alpha_x$$

Key Idea: Inversion About the Mean

The diffusion operator D performs **inversion about the mean**: each amplitude is reflected around the average amplitude $\bar{\alpha}$.

- If $\alpha_x < \bar{\alpha}$: the amplitude *increases*.
- If $\alpha_x > \bar{\alpha}$: the amplitude *decreases*.

After the oracle flips the target's amplitude to $-|\alpha_\omega|$, the target amplitude is far below the mean. Inversion about the mean then pushes it significantly upward.

3.3 The Grover Iterate

Definition: Grover Iterate G

$$G = D \cdot U_f = (2|s\rangle\langle s| - I)(I - 2|\omega\rangle\langle\omega|)$$

One application of G is called a **Grover step** or **Grover iterate**.

3.4 Circuit Diagrams for U_f and D

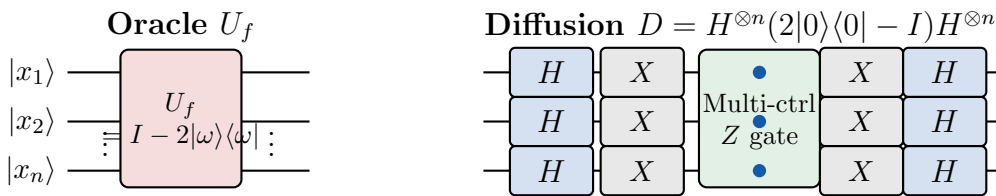


Figure 2: Left: phase oracle U_f . Right: decomposition of the diffusion operator D . The $(2|0\rangle\langle 0| - I)$ operation is a phase flip on $|0 \cdots 0\rangle$, implemented with X gates and a multi-controlled- Z (equivalently, multi-controlled- X with a phase kickback ancilla).

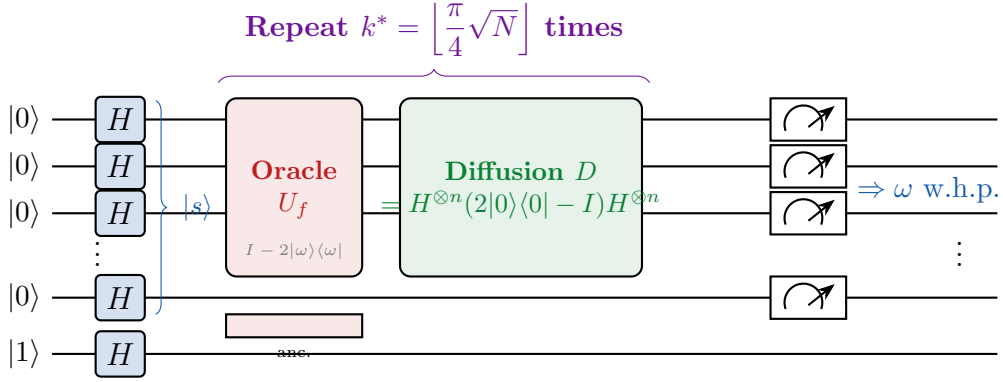


Figure 3: Full Grover circuit. The Grover iterate $G = D \cdot U_f$ is applied $k^* = \lfloor \frac{\pi}{4} \sqrt{N} \rfloor$ times. Final measurement of the data register yields the target ω with probability close to 1.

4 Geometric Analysis

This section gives the cleanest and most intuitive explanation of Grover's algorithm. The key insight is that the entire N -dimensional evolution can be understood as rotation in a **2-dimensional plane**.

4.1 The 2D Invariant Subspace

Define two orthonormal-in-span vectors:

$$|\omega\rangle \quad (\text{the target state}) \quad (3)$$

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} |x\rangle \quad (\text{uniform superposition over non-targets}) \quad (4)$$

Note $\langle s' | \omega \rangle = 0$. We can write the initial state $|s\rangle$ as:

$$|s\rangle = \underbrace{\frac{1}{\sqrt{N}}}_{\sin \theta} |\omega\rangle + \underbrace{\sqrt{\frac{N-1}{N}}}_{\cos \theta} |s'\rangle$$

where

$$\sin \theta = \frac{1}{\sqrt{N}}, \quad \theta = \arcsin\left(\frac{1}{\sqrt{N}}\right)$$

For large N : $\sin \theta \approx \theta \approx 1/\sqrt{N}$ (small angle).

Theorem: Invariance of the 2D Plane

Both U_f and D preserve the subspace $\mathcal{V} = \text{span}\{|\omega\rangle, |s'\rangle\}$.

Proof:

- $U_f = I - 2|\omega\rangle\langle\omega|$ maps $|\omega\rangle \mapsto -|\omega\rangle$ and $|s'\rangle \mapsto |s'\rangle$, so $\mathcal{V}$ is preserved.
- $D = 2|s\rangle\langle s| - I$: since $|s\rangle \in \mathcal{V}$, D maps any vector in $\mathcal{V}$ to a vector in $\mathcal{V}$. $\square$

4.2 What U_f Does Geometrically

In the 2D picture with horizontal axis $|s'\rangle$ and vertical axis $|\omega\rangle$:

$U_f = I - 2|\omega\rangle\langle\omega|$ **reflects** the state vector about the horizontal axis $|s'\rangle$ (it negates the $|\omega\rangle$ component).

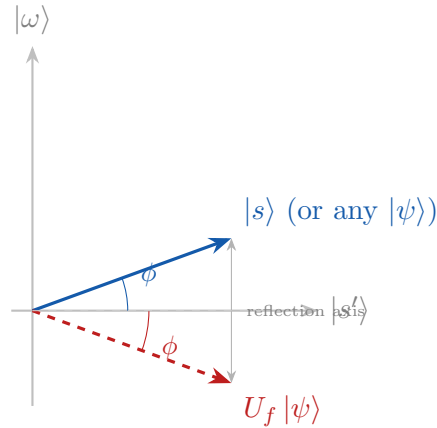


Figure 4: U_f reflects the state vector about the $|s'\rangle$ axis (horizontal). The angle below the horizontal equals the angle above. The $|\omega\rangle$ component changes sign.

4.3 What D Does Geometrically

$D = 2|s\rangle\langle s| - I$ **reflects** the state vector about the $|s\rangle$ axis.

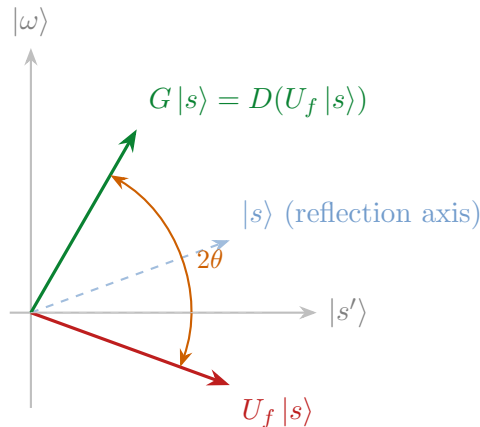


Figure 5: After U_f rotates to angle $-\theta$, the diffusion D reflects about $|s\rangle$ (at angle $+\theta$), bringing the state to angle $+3\theta$. Net rotation: $\theta \rightarrow 3\theta$, an increase of 2θ .

4.4 Full Grover Iterate: Rotation by 2θ

Theorem: Rotation Theorem

Starting from a state at angle ϕ from $|s'\rangle$ (i.e., $|\psi\rangle = \sin \phi |\omega\rangle + \cos \phi |s'\rangle$), one application of $G = D \cdot U_f$ produces:

$$G|\psi\rangle = \sin(\phi + 2\theta) |\omega\rangle + \cos(\phi + 2\theta) |s'\rangle$$

The state rotates by exactly 2θ toward $|\omega\rangle$ in the $\{|\omega\rangle, |s'\rangle\}$ plane.

Proof.

1. Start: $|\psi\rangle = \sin \phi |\omega\rangle + \cos \phi |s'\rangle$. Angle from $|s'\rangle$ is ϕ .

2. After U_f (reflect about $|s'\rangle$): negate $|\omega\rangle$ component:

$$U_f|\psi\rangle = -\sin \phi |\omega\rangle + \cos \phi |s'\rangle. \quad \text{Angle: } -\phi.$$

3. After D (reflect about $|s\rangle$ at angle θ): a reflection of angle $-\phi$ about the line at angle θ gives angle $2\theta - (-\phi) = 2\theta + \phi$.

$$G|\psi\rangle = \sin(2\theta + \phi) |\omega\rangle + \cos(2\theta + \phi) |s'\rangle. \quad \square$$

4.5 State After k Iterations

Starting from $|s\rangle$ at initial angle θ :

$$G^k |s\rangle = \sin((2k+1)\theta) |\omega\rangle + \cos((2k+1)\theta) |s'\rangle$$

4.6 Optimal Number of Iterations

Success probability after k steps:

$$P_{\text{success}}(k) = \sin^2((2k+1)\theta)$$

This is maximised when $(2k+1)\theta \approx \pi/2$, i.e.,

$$k^* = \left\lfloor \frac{\pi/2 - \theta}{2\theta} \right\rfloor = \left\lfloor \frac{\pi}{4\theta} - \frac{1}{2} \right\rfloor \approx \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor$$

(using $\theta \approx 1/\sqrt{N}$ for large N).

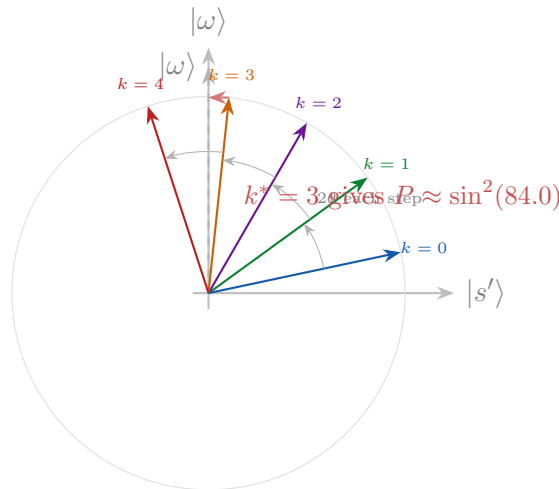


Figure 6: Geometric view: the state rotates by 2θ per Grover iterate, spiralling toward $|\omega\rangle$. The optimal stopping point k^* is when the angle reaches ≈ 90 from $|s'\rangle$.

5 Algebraic Analysis

We now derive the same results purely algebraically and explicitly connect to the geometric picture.

5.1 Amplitude Recurrence

Write the state after k iterates as:

$$|\psi_k\rangle = \alpha_k |\omega\rangle + \beta_k \sum_{x \neq \omega} |x\rangle$$

(by symmetry, all non-target amplitudes remain equal throughout).

Initial values: $\alpha_0 = \beta_0 = 1/\sqrt{N}$.

Step A: Apply U_f

$$U_f |\psi_k\rangle = -\alpha_k |\omega\rangle + \beta_k \sum_{x \neq \omega} |x\rangle$$

Step B: Compute Mean Before Diffusion

$$\bar{\mu} = \frac{-\alpha_k + (N-1)\beta_k}{N}$$

Step C: Apply D (Inversion About Mean)

$$\alpha_{k+1} = 2\bar{\mu} - (-\alpha_k) = 2\bar{\mu} + \alpha_k = \frac{-2\alpha_k + 2(N-1)\beta_k}{N} + \alpha_k = \frac{(N-2)\alpha_k + 2(N-1)\beta_k}{N} \quad (5)$$

$$\beta_{k+1} = 2\bar{\mu} - \beta_k = \frac{-2\alpha_k + 2(N-1)\beta_k}{N} - \beta_k = \frac{-2\alpha_k + (N-2)\beta_k}{N} \quad (6)$$

In matrix form:

$$\begin{pmatrix} \alpha_{k+1} \\ \beta_{k+1} \end{pmatrix} = \underbrace{\frac{1}{N} \begin{pmatrix} N-2 & 2(N-1) \\ -2 & N-2 \end{pmatrix}}_M \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$$

5.2 Closed-Form Solution

Theorem: Amplitude Formula

The exact solution to the recurrence with $\alpha_0 = \beta_0 = 1/\sqrt{N}$ is:

$$\alpha_k = \sin((2k+1)\theta) \quad (7)$$

$$\beta_k = \frac{\cos((2k+1)\theta)}{\sqrt{N-1}} \quad (8)$$

where $\theta = \arcsin(1/\sqrt{N})$.

Verification:

- **Initial condition** ($k = 0$): $\alpha_0 = \sin \theta = 1/\sqrt{N} \checkmark$ $\beta_0 = \cos \theta / \sqrt{N-1} = \sqrt{(N-1)/N} / \sqrt{N-1} = 1/\sqrt{N} \checkmark$
- **Normalisation**: $|\alpha_k|^2 + (N-1)|\beta_k|^2 = \sin^2((2k+1)\theta) + \cos^2((2k+1)\theta) = 1 \checkmark$
- **Recurrence check (sketch)**: substituting into (5) and using $\sin(A+2\theta) = \sin A \cos 2\theta + \cos A \sin 2\theta$ with $\cos 2\theta = 1 - 2/N$ and $\sin 2\theta = 2\sqrt{N-1}/N$ confirms the formula. $\square$

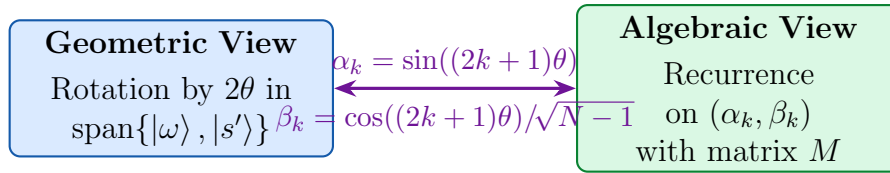
5.3 Equivalence with the Geometric Picture

The geometric result was:

$$G^k |s\rangle = \sin((2k+1)\theta) |\omega\rangle + \cos((2k+1)\theta) |s'\rangle$$

Since $|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} |x\rangle$, the amplitude of each non-target state in the geometric formula is $\cos((2k+1)\theta)/\sqrt{N-1}$, which exactly matches the algebraic β_k in (8).

The geometric picture is a faithful, lower-dimensional representation of the algebraic recurrence.

**6 Worked Examples****6.1 Example 1: $N = 4$ ($n = 2$ qubits), $\omega = 11$**

Setup: $N = 4$, $\theta = \arcsin(1/2) = 30^\circ$, $k^* = \lfloor \frac{\pi}{4} \cdot 2 \rfloor = 1$.

Initial state after $H^{\otimes 2}$:

$$|s\rangle = \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle$$

After Oracle U_f (flip sign of $|11\rangle$):

$$\frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

Mean amplitude:

$$\bar{\alpha} = \frac{1}{4} \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2} \right) = \frac{1}{4}$$

After Diffusion ($\alpha'_x = 2\bar{\alpha} - \alpha_x$):

$$\begin{aligned} |00\rangle &: 2 \cdot \frac{1}{4} - \frac{1}{2} = 0 \\ |01\rangle &: 2 \cdot \frac{1}{4} - \frac{1}{2} = 0 \\ |10\rangle &: 2 \cdot \frac{1}{4} - \frac{1}{2} = 0 \\ |11\rangle &: 2 \cdot \frac{1}{4} - \left(-\frac{1}{2}\right) = 1 \end{aligned}$$

$$|\psi_1\rangle = |11\rangle \Rightarrow P(\text{success}) = 1^2 = \mathbf{100\%}!$$

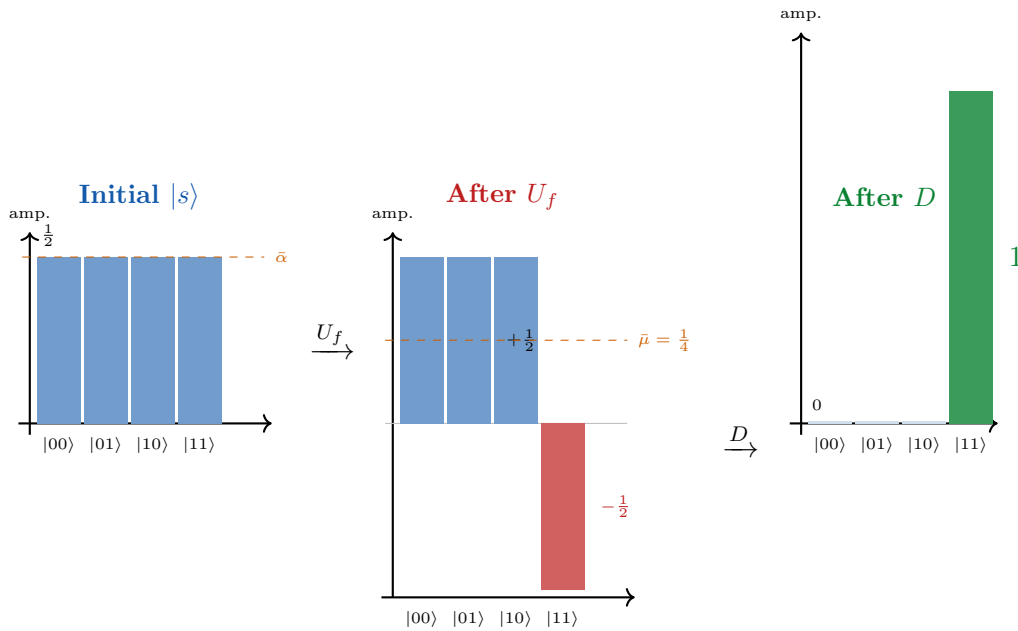


Figure 7: Amplitude evolution for $N = 4$, $\omega = 11$. Left: uniform superposition. Middle: oracle flips the target amplitude to $-\frac{1}{2}$, pushing it below the new mean $\bar{\mu} = \frac{1}{4}$. Right: inversion about the mean sends the target amplitude to 1 and all others to 0. One iteration suffices!

Example: N

$$k^* = 1, (2 \cdot 1 + 1)\theta = 3 \cdot 30 = 90.$$

$$\alpha_1 = \sin(90) = 1, \quad P = \sin^2(90) = 100\%. \quad \checkmark$$

6.2 Example 2: $N = 8$ ($n = 3$ qubits), $\omega = 101$

Setup: $N = 8$, $\theta = \arcsin(1/\sqrt{8}) \approx 20.7$, $k^* = \lfloor \frac{\pi}{4}\sqrt{8} \rfloor = \lfloor 2.22 \rfloor = 2$.

k	$(2k+1)\theta$	$\alpha_k = \sin(\cdot)$	$\beta_k = \cos(\cdot)/\sqrt{7}$	$P_k = \alpha_k^2$	Check $\alpha^2 + 7\beta^2$
0	20.7	0.354	0.354	12.5%	1.000✓
1	62.1	0.884	0.176	78.1%	1.000✓
2	103.5	0.972	-0.087	94.5%	1.000✓
3	144.9	0.574	0.312	32.9% ←over!	1.000✓

Table 1: Amplitude evolution for $N = 8$. Stopping at $k^* = 2$ gives 94.5% success. Applying one more step ($k = 3$) causes over-rotation and the probability drops dramatically.

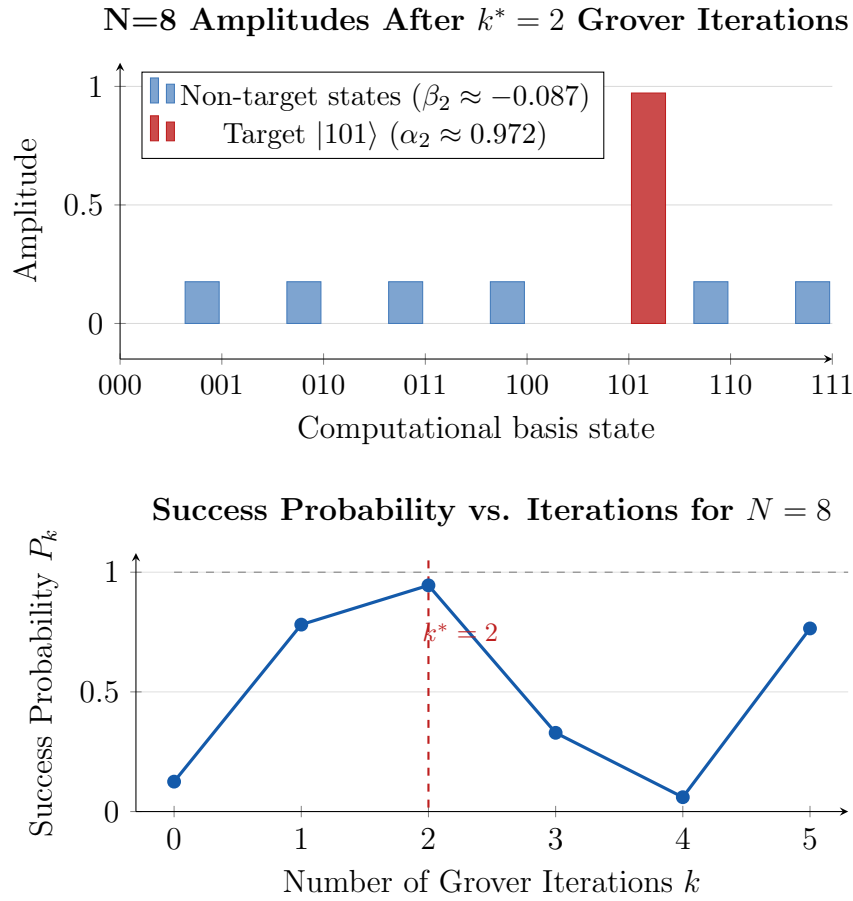


Figure 8: Top: amplitude bar chart after $k^* = 2$ iterations. Bottom: oscillating success probability — stopping at $k^* = 2$ gives the first near-maximum.

6.3 Example 3: $N = 16$ ($n = 4$ qubits)

$\theta = \arcsin(1/4) \approx 14.48$, $k^* = \lfloor \frac{\pi}{4} \cdot 4 \rfloor = 3$.

k	$\phi = (2k+1)\theta$	α_k	$\beta_k = \cos \phi / \sqrt{15}$	P_k
0	14.5	0.250	0.250	6.3%
1	43.4	0.687	0.187	47.2%
2	72.3	0.953	0.079	90.8%
3	101.2	0.981	-0.051	96.2%
4	130.1	0.763	-0.175	58.2%

Table 2: $N = 16$: optimal at $k^* = 3$. At $k = 4$ the probability drops — clear evidence of over-rotation.

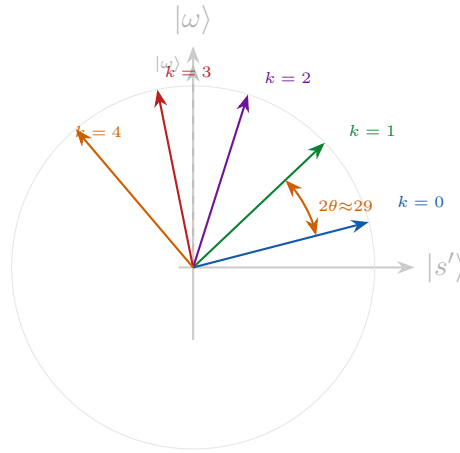


Figure 9: $N = 16$ geometric view. Each Grover step rotates by ≈ 29 . At $k = 3$ the state is closest to $|\omega\rangle$ (96.2%). At $k = 4$ it overshoots.

6.4 Example 4: $N = 64$ ($n = 6$ qubits) — Success Probability Curve

$\theta = \arcsin(1/8) \approx 7.18$, $k^* = \lfloor \frac{\pi}{4} \cdot 8 \rfloor = 6$.

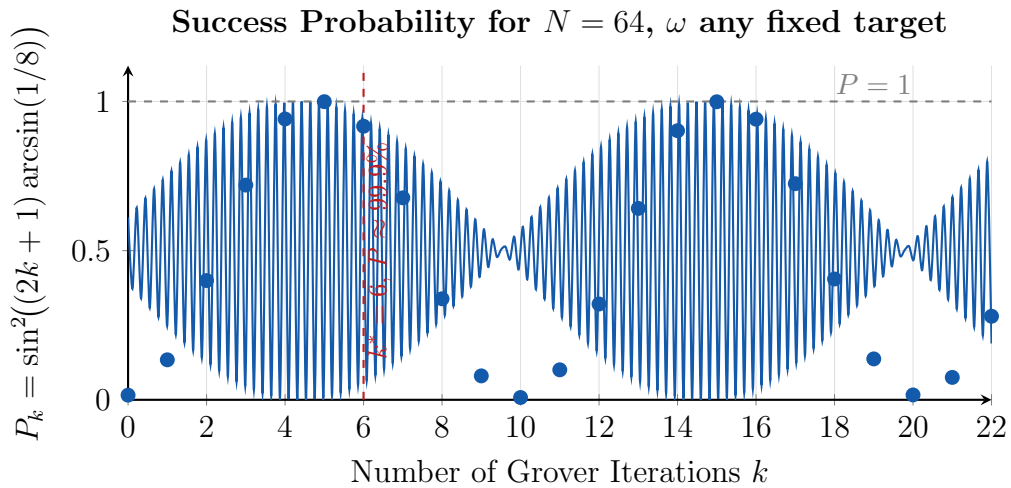


Figure 10: The success probability oscillates sinusoidally. The first maximum occurs at $k^* = 6$ iterations with $P \approx 99.9\%$. Note the characteristic periodic over/under-rotation pattern.

6.5 Example 5: Multiple Targets ($M > 1$)

If there are M marked items ($M < N/2$), the analysis extends with:

$$\sin \theta_M = \sqrt{\frac{M}{N}}, \quad k_M^* \approx \frac{\pi}{4} \sqrt{\frac{N}{M}}, \quad \text{speedup over classical: } \frac{N/M}{k_M^*} \approx \frac{4}{\pi} \sqrt{\frac{N}{M}}$$

Example: N

$$\theta_M = \arcsin \sqrt{\frac{4}{256}} = \arcsin \frac{1}{8} \approx 7.18, \quad k^* \approx \frac{\pi}{4} \sqrt{\frac{256}{4}} = \frac{\pi}{4} \cdot 8 \approx 6 \text{ steps.}$$

Classical average: $256/(2 \cdot 4) = 32$ steps. Quantum: ≈ 6 steps. **Speedup:** $\approx 5.3\times$.

7 Full Algorithm Statement

Grover's Search Algorithm

Input: Oracle U_f for $f : \{0, 1\}^n \rightarrow \{0, 1\}$ with unique solution ω ; $N = 2^n$.

Output: ω with probability $\geq 1 - \mathcal{O}(1/N)$.

Step 1. Initialise. Prepare $|0\rangle^{\otimes n} |1\rangle$ (data + ancilla).

Step 2. Superpose. Apply $H^{\otimes n}$ to data and H to ancilla:

$$|s\rangle |-\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle |-\rangle$$

Step 3. Iterate k^* times. For $j = 1, 2, \dots, k^*$:

(a) Apply U_f : phase-flip the target. (*Oracle query*)

(b) Apply $D = H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n}$: diffusion.

where $k^* = \lfloor \frac{\pi}{4} \sqrt{N} \rfloor$.

Step 4. Measure the data register. Output the result.

Query complexity: $\Theta(\sqrt{N})$. **Gate complexity:** $\mathcal{O}(n\sqrt{N})$.

8 Complexity and Optimality

8.1 Query Complexity

Theorem: Query Complexity of Grover's Algorithm

Grover's algorithm finds the unique marked element with probability $\geq 1 - \delta$ using

$$Q = \mathcal{O}\left(\sqrt{N} \log(1/\delta)\right)$$

queries to the oracle. For constant success probability, $Q = \Theta(\sqrt{N})$.

8.2 Classical Lower Bound

Theorem: Classical Lower Bound (trivial)

Any deterministic or randomised classical algorithm for unstructured search requires $\Omega(N)$ oracle queries in the worst case.

Proof sketch: After $k < N$ queries on an N -element space, at most k items have

been checked. The adversary can always place the target among the unchecked items. $\square$

8.3 Quantum Lower Bound (Grover is Optimal!)

Theorem: Quantum Lower Bound (Bennett–Bernstein–Brassard–Vazirani)

Any quantum algorithm for unstructured search requires $\Omega(\sqrt{N})$ oracle queries.

Proof idea (polynomial method): After k quantum queries, the amplitude of the target state is a degree- $2k$ polynomial in the oracle parameters. For this polynomial to have $\Omega(1)$ magnitude, we need $k = \Omega(\sqrt{N})$ by polynomial approximation theory. $\square$

Thus Grover's algorithm is **exactly optimal** in the query complexity model.

8.4 Summary Table

Model	Queries (unique solution)	Queries (M solutions)	Optimal?
Classical (worst)	N	N/M	Yes
Classical (average)	$N/2$	$N/(2M)$	Yes
Quantum (Grover)	$\frac{\pi}{4}\sqrt{N}$	$\frac{\pi}{4}\sqrt{N/M}$	Yes (proven)

Table 3: Query complexity comparison. Grover achieves a tight $\Theta(\sqrt{N/M})$ speedup.

9 Common Pitfalls and Practical Notes

Key Idea: Pitfall 1 — Over-Rotation

Applying *more* than k^* iterations **decreases** the success probability. The state over-rotates past $|\omega\rangle$. Always compute $k^* = \lfloor \frac{\pi}{4}\sqrt{N/M} \rfloor$ carefully.

Key Idea: Pitfall 2 — Unknown M

If the number of solutions M is not known, use **Quantum Counting** (phase estimation on G) to estimate M , then apply Grover. Alternatively, a randomised strategy: pick k uniformly in $[1, \sqrt{N}]$ and repeat $O(\log(1/\delta))$ times, achieving $O(\sqrt{N})$ expected queries.

Key Idea: Pitfall 3 — Not a Speedup for All Problems

Grover accelerates search and any problem reducible to search by a quadratic factor. It does *not* provide exponential speedup. Problems with classical $O(\text{poly}(n))$ algorithms do not become fundamentally easier.

Note

Verification. After the quantum circuit outputs a candidate x , verify classically in $O(1)$ by evaluating $f(x)$. If wrong (probability $\approx 1/N$), repeat the entire procedure.

9.1 Gate Count for the Diffusion Operator

The multi-controlled phase flip $(2|0\rangle\langle 0| - I)$ on n qubits can be decomposed as:

$$2|0\rangle\langle 0| - I = X^{\otimes n} \cdot C^{n-1}Z \cdot X^{\otimes n}$$

where $C^{n-1}Z$ is an $(n-1)$ -controlled Z gate. This requires $O(n)$ gates using ancilla-based decompositions. Thus each diffusion step uses $O(n)$ elementary gates, and the total gate count of Grover's algorithm is:

$$\text{Total gates} = k^* \times (O(n) + \text{cost}(U_f)) \approx \frac{\pi}{4} \sqrt{N} \cdot O(n)$$

10 Generalisation: Amplitude Amplification

Grover's algorithm is a special case of the much more general **Amplitude Amplification** framework due to Brassard, Høyer, Mosca, and Tapp (2002).

Definition: Amplitude Amplification

Let A be any quantum algorithm on $n + m$ qubits. Say a basis state $|x, y\rangle$ is **good** if $f(x) = 1$. Let $a = \|P_{\text{good}} A |0\rangle\|^2$ be the probability that A produces a good state. Define:

$$Q = -AS_0A^\dagger S_{\text{good}}$$

where $S_0 = I - 2|0^{n+m}\rangle\langle 0^{n+m}|$ and S_{good} flips the phase of good states. Then k applications of Q give success probability $\sin^2((2k+1) \arcsin \sqrt{a})$.

Grover's algorithm is the special case $A = H^{\otimes n}$, giving $a = M/N$.

Consequence: If any classical algorithm solves a problem with probability p , its quantum analogue (using amplitude amplification) solves it in $O(1/\sqrt{p})$ quantum steps instead of $O(1/p)$ classical steps — always a quadratic speedup.

11 Summary

Lecture Summary

1. **Problem:** Unstructured search over $N = 2^n$ items. Classical: $\Theta(N)$; Quantum: $\Theta(\sqrt{N})$.
2. **Oracle:** Phase oracle $U_f = I - 2|\omega\rangle\langle \omega|$ phase-flips only the target $|\omega\rangle$. Built via *phase kick-back* from the standard oracle.
3. **Diffusion:** $D = 2|s\rangle\langle s| - I$ inverts amplitudes about their mean. Implemented as $H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n}$.

4. **Geometric view:** $G = D \cdot U_f$ rotates the state by 2θ per step inside the 2D plane $\text{span}\{|\omega\rangle, |s'\rangle\}$.
5. **Algebraic view:** Amplitudes satisfy $\alpha_k = \sin((2k+1)\theta)$, $\beta_k = \cos((2k+1)\theta)/\sqrt{N-1}$. Both views are equivalent.
6. **Optimal iterations:** $k^* \approx \frac{\pi}{4}\sqrt{N/M}$ gives $P(\text{success}) \approx 1$.
7. **Optimality:** Proven $\Omega(\sqrt{N})$ lower bound; Grover is tight.

Review Questions

1. Derive the phase kick-back trick from first principles. Why must the ancilla be in $|-\rangle$?
2. Show algebraically that $D = 2|s\rangle\langle s| - I$ performs inversion about the mean on the amplitudes.
3. For $N = 256$ with one target, compute k^* and the exact success probability P_{k^*} .
4. Explain geometrically *why* over-rotation reduces the success probability.
5. Prove that the matrix M in the amplitude recurrence has eigenvalues $e^{\pm 2i\theta}$.
6. For $N = 1024$, $M = 16$ targets: how many Grover iterations are needed? What is the speedup over classical?
7. Describe the amplitude amplification framework and state how it generalises Grover's algorithm.
8. Why does Grover's algorithm give only a *quadratic* speedup, not an exponential one? What is fundamentally different from Shor's algorithm?

Further Reading

- L. K. Grover, "A fast quantum mechanical algorithm for database search," *Proceedings of ACM STOC*, 1996.
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